

Gaussian quadrature

In numerical analysis, a **quadrature rule** is an approximation of the definite integral of a function, usually stated as a weighted sum of function values at specified points within the domain of integration. (See [numerical integration](#) for more on quadrature rules.) An *n*-point **Gaussian quadrature rule**, named after Carl Friedrich Gauss,^[1] is a quadrature rule constructed to yield an exact result for polynomials of degree *2n* − 1 or less by a suitable choice of the nodes *x*_{*i*} and weights *w*_{*i*} for *i* = 1, ..., *n*. The modern formulation using orthogonal polynomials was developed by Carl Gustav Jacobi 1826.^[2] The most common domain of integration for such a rule is taken as [−1, 1], so the rule is stated as

$$\int_{-1}^1 f(x) \, dx \approx \sum_{i=1}^n w_i f(x_i),$$

which is exact for polynomials of degree *2n* − 1 or less. This exact rule is known as the Gauss-Legendre quadrature rule. The quadrature rule will only be an accurate approximation to the integral above if *f*(*x*) is well-approximated by a polynomial of degree *2n* − 1 or less on [−1, 1].

The Gauss-Legendre quadrature rule is not typically used for integrable functions with endpoint [singularities](#). Instead, if the integrand can be written as

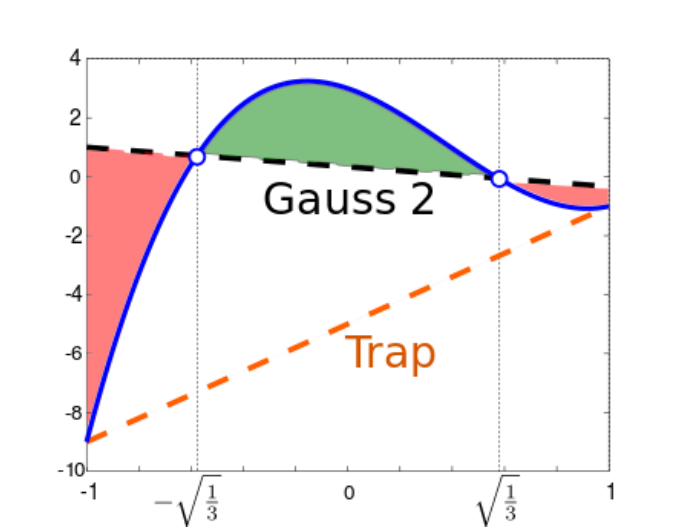
$$f(x) = (1 - x)^{\alpha}(1 + x)^{\beta}g(x), \quad \alpha, \beta > -1,$$

where *g*(*x*) is well-approximated by a low-degree polynomial, then alternative nodes *x*_{*i*}['] and weights *w*_{*i*}['] will usually give more accurate quadrature rules. These are known as [Gauss-Jacobi quadrature rules](#), i.e.,

$$\int_{-1}^1 f(x) \, dx = \int_{-1}^1 (1 - x)^{\alpha}(1 + x)^{\beta}g(x) \, dx \approx \sum_{i=1}^n w_i' g(x_i').$$

Common weights include $\frac{1}{\sqrt{1-x^2}}$ ([Chebyshev–Gauss](#)) and $\sqrt{1-x^2}$. One may also want to integrate over semi-infinite ([Gauss-Laguerre quadrature](#)) and infinite intervals ([Gauss–Hermite quadrature](#)).

It can be shown (see Press, et al., or Stoer and Bulirsch) that the quadrature nodes *x*_{*i*} are the [roots](#) of a polynomial belonging to a class of [orthogonal polynomials](#) (the class orthogonal with respect to a weighted inner-product). This is a key observation for computing Gauss quadrature nodes and weights.



Comparison between 2-point Gaussian and trapezoidal quadrature. The blue line is the polynomial *y*(*x*) = 7*x*³ − 8*x*² − 3*x* + 3, whose integral in [−1, 1] is ²/₃. The [trapezoidal rule](#) returns the integral of the orange dashed line, equal to *y*(−1) + *y*(1) = −10. The 2-point Gaussian quadrature rule returns the integral of the black dashed curve, equal to *y*(− $\sqrt{\frac{1}{3}}$) + *y*($\sqrt{\frac{1}{3}}$) = ²/₃. Such a result is exact, since the green region has the same area as the sum of the red regions.

Contents

[Gauss–Legendre quadrature](#)

[Change of interval](#)

[Other forms](#)

[Fundamental theorem](#)

[General formula for the weights](#)

[Proof that the weights are positive](#)

[Computation of Gaussian quadrature rules](#)

[Recurrence relation](#)

[The Golub-Welsch algorithm](#)

[Error estimates](#)

[Gauss–Kronrod rules](#)

[Gauss–Lobatto rules](#)

[References](#)

[External links](#)

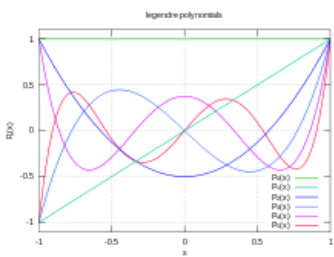
Gauss–Legendre quadrature

For the simplest integration problem stated above, i.e., $f(x)$ is well-approximated by polynomials on $[-1, 1]$, the associated orthogonal polynomials are Legendre polynomials, denoted by $P_n(x)$. With the n -th polynomial normalized to give $P_n(1) = 1$, the i -th Gauss node, x_i , is the i -th root of P_n and the weights are given by the formula (Abramowitz & Stegun 1972, p. 887)

$$w_i = \frac{2}{(1 - x_i^2) [P'_n(x_i)]^2}.$$

Some low-order quadrature rules are tabulated below (over interval $[-1, 1]$, see the section below for other intervals).

Number of points, n	Points, x_i		Weights, w_i	
1	0		2	
2	$\pm \frac{1}{\sqrt{3}}$	$\pm 0.57735\dots$	1	
3	0		$\frac{8}{9}$	0.888889...
	$\pm \sqrt{\frac{3}{5}}$	$\pm 0.774597\dots$	$\frac{5}{9}$	0.555556...
4	$\pm \sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}}}$	$\pm 0.339981\dots$	$\frac{18 + \sqrt{30}}{36}$	0.652145...
	$\pm \sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}}}$	$\pm 0.861136\dots$	$\frac{18 - \sqrt{30}}{36}$	0.347855...
5	0		$\frac{128}{225}$	0.568889...
	$\pm \frac{1}{3}\sqrt{5 - 2\sqrt{\frac{10}{7}}}$	$\pm 0.538469\dots$	$\frac{322 + 13\sqrt{70}}{900}$	0.478629...
	$\pm \frac{1}{3}\sqrt{5 + 2\sqrt{\frac{10}{7}}}$	$\pm 0.90618\dots$	$\frac{322 - 13\sqrt{70}}{900}$	0.236927...



Graphs of Legendre polynomials (up to $n = 5$)

Change of interval

An integral over $[a, b]$ must be changed into an integral over $[-1, 1]$ before applying the Gaussian quadrature rule. This change of interval can be done in the following way:

$$\int_a^b f(x) \, dx = \frac{b-a}{2} \int_{-1}^1 f\left(\frac{b-a}{2}\xi + \frac{a+b}{2}\right) \, d\xi.$$

Applying n point Gaussian quadrature (ξ, w) rule then results in the following approximation:

$$\int_a^b f(x) \, dx \approx \frac{b-a}{2} \sum_{i=1}^n w_i f\left(\frac{b-a}{2}\xi_i + \frac{a+b}{2}\right).$$

Other forms

The integration problem can be expressed in a slightly more general way by introducing a positive weight function ω into the integrand, and allowing an interval other than $[-1, 1]$. That is, the problem is to calculate

$$\int_a^b \omega(x) f(x) \, dx$$

for some choices of a, b , and ω . For $a = -1, b = 1$, and $\omega(x) = 1$, the problem is the same as that considered above. Other choices lead to other integration rules. Some of these are tabulated below. Equation numbers are given for Abramowitz and Stegun (A & S).

Interval	$\omega(x)$	Orthogonal polynomials	A & S	For more information, see ...
$[-1, 1]$	1	Legendre polynomials	25.4.29	§ Gauss–Legendre quadrature
$(-1, 1)$	$(1-x)^\alpha(1+x)^\beta, \quad \alpha, \beta > -1$	Jacobi polynomials	25.4.33 ($\beta = 0$)	Gauss–Jacobi quadrature
$(-1, 1)$	$\frac{1}{\sqrt{1-x^2}}$	Chebyshev polynomials (first kind)	25.4.38	Chebyshev–Gauss quadrature
$[-1, 1]$	$\sqrt{1-x^2}$	Chebyshev polynomials (second kind)	25.4.40	Chebyshev–Gauss quadrature
$[0, \infty)$	e^{-x}	Laguerre polynomials	25.4.45	Gauss–Laguerre quadrature
$[0, \infty)$	$x^\alpha e^{-x}, \quad \alpha > -1$	Generalized Laguerre polynomials		Gauss–Laguerre quadrature
$(-\infty, \infty)$	e^{-x^2}	Hermite polynomials	25.4.46	Gauss–Hermite quadrature

Fundamental theorem

Let p_n be a nontrivial polynomial of degree n such that

$$\int_a^b \omega(x) x^k p_n(x) dx = 0, \quad \text{for all } k = 0, 1, \dots, n-1.$$

If we pick the n nodes x_i to be the zeros of p_n , then there exist n weights w_i which make the Gauss-quadrature computed integral exact for all polynomials $h(x)$ of degree $2n-1$ or less. Furthermore, all these nodes x_i will lie in the open interval (a, b) ([Stoer & Bulirsch 2002](#), pp. 172–175).

The polynomial p_n is said to be an orthogonal polynomial of degree n associated to the weight function $\omega(x)$. It is unique up to a constant normalization factor. The idea underlying the proof is that, because of its sufficiently low degree, $h(x)$ can be divided by $p_n(x)$ to produce a quotient $q(x)$ of degree strictly lower than n , and a remainder $r(x)$ of still lower degree, so that both will be orthogonal to $p_n(x)$, by the defining property of $p_n(x)$. Thus

$$\int_a^b \omega(x) h(x) dx = \int_a^b \omega(x) r(x) dx.$$

Because of the choice of nodes x_i , the corresponding relation

$$\sum_{i=1}^n w_i h(x_i) = \sum_{i=1}^n w_i r(x_i)$$

holds also. The exactness of the computed integral for $h(x)$ then follows from corresponding exactness for polynomials of degree only n or less (as is $r(x)$).

General formula for the weights

The weights can be expressed as

$$w_i = \frac{a_n}{a_{n-1}} \frac{\int_a^b \omega(x) p_{n-1}(x)^2 dx}{p_n'(x_i) p_{n-1}(x_i)} \quad (1)$$

where a_k is the coefficient of x^k in $p_k(x)$. To prove this, note that using [Lagrange interpolation](#) one can express $r(x)$ in terms of $r(x_i)$ as

$$r(x) = \sum_{i=1}^n r(x_i) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}$$

because $r(x)$ has degree less than n and is thus fixed by the values it attains at n different points. Multiplying both sides by $\omega(x)$ and integrating from a to b yields

$$\int_a^b \omega(x) r(x) dx = \sum_{i=1}^n r(x_i) \int_a^b \omega(x) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j} dx$$

The weights w_i are thus given by

$$w_i = \int_a^b \omega(x) \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j} dx$$

This integral expression for w_i can be expressed in terms of the orthogonal polynomials $p_n(x)$ and $p_{n-1}(x)$ as follows.

We can write

$$\prod_{\substack{1 \leq j \leq n \\ j \neq i}} (x - x_j) = \frac{\prod_{1 \leq j \leq n} (x - x_j)}{x - x_i} = \frac{p_n(x)}{a_n (x - x_i)}$$

where a_n is the coefficient of x^n in $p_n(x)$. Taking the limit of x to x_i yields using L'Hôpital's rule

$$\prod_{\substack{1 \leq j \leq n \\ j \neq i}} (x_i - x_j) = \frac{p'_n(x_i)}{a_n}$$

We can thus write the integral expression for the weights as

$$w_i = \frac{1}{p'_n(x_i)} \int_a^b \omega(x) \frac{p_n(x)}{x - x_i} dx \quad (2)$$

In the integrand, writing

$$\frac{1}{x - x_i} = \frac{1 - \left(\frac{x}{x_i}\right)^k}{x - x_i} + \left(\frac{x}{x_i}\right)^k \frac{1}{x - x_i}$$

yields

$$\int_a^b \omega(x) \frac{x^k p_n(x)}{x - x_i} dx = x_i^k \int_a^b \omega(x) \frac{p_n(x)}{x - x_i} dx$$

provided $k \leq n$, because

$$\frac{1 - \left(\frac{x}{x_i}\right)^k}{x - x_i}$$

is a polynomial of degree $k - 1$ which is then orthogonal to $p_n(x)$. So, if $q(x)$ is a polynomial of at most n th degree we have

$$\int_a^b \omega(x) \frac{p_n(x)}{x - x_i} dx = \frac{1}{q(x_i)} \int_a^b \omega(x) \frac{q(x) p_n(x)}{x - x_i} dx$$

We can evaluate the integral on the right hand side for $q(x) = p_{n-1}(x)$ as follows. Because $\frac{p_n(x)}{x - x_i}$ is a polynomial of degree $n - 1$, we have

$$\frac{p_n(x)}{x - x_i} = a_n x^{n-1} + s(x)$$

where $s(x)$ is a polynomial of degree $n - 2$. Since $s(x)$ is orthogonal to $p_{n-1}(x)$ we have

$$\int_a^b \omega(x) \frac{p_n(x)}{x - x_i} dx = \frac{a_n}{p_{n-1}(x_i)} \int_a^b \omega(x) p_{n-1}(x) x^{n-1} dx$$

We can then write

$$x^{n-1} = \left(x^{n-1} - \frac{p_{n-1}(x)}{a_{n-1}} \right) + \frac{p_{n-1}(x)}{a_{n-1}}$$

The term in the brackets is a polynomial of degree $n - 2$, which is therefore orthogonal to $p_{n-1}(x)$. The integral can thus be written as

$$\int_a^b \omega(x) \frac{p_n(x)}{x - x_i} dx = \frac{a_n}{a_{n-1}p_{n-1}(x_i)} \int_a^b \omega(x) p_{n-1}(x)^2 dx$$

According to equation (2), the weights are obtained by dividing this by $p'_n(x_i)$ and that yields the expression in equation (1).

w_i can also be expressed in terms of the orthogonal polynomials $p_n(x)$ and now $p_{n+1}(x)$. In the 3-term recurrence relation $p_{n+1}(x_i) = (a)p_n(x_i) + (b)p_{n-1}(x_i)$ the term with $p_n(x_i)$ vanishes, so $p_{n-1}(x_i)$ in Eq. (1) can be replaced by $\frac{1}{b}p_{n+1}(x_i)$.

Proof that the weights are positive

Consider the following polynomial of degree $2n - 2$

$$f(x) = \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{(x - x_j)^2}{(x_i - x_j)^2}$$

where, as above, the x_j are the roots of the polynomial $p_n(x)$. Clearly $f(x_j) = \delta_{ij}$. Since the degree of $f(x)$ is less than $2n - 1$, the Gaussian quadrature formula involving the weights and nodes obtained from $p_n(x)$ applies. Since $f(x_j) = 0$ for j not equal to i , we have

$$\int_a^b \omega(x) f(x) dx = \sum_{j=1}^N w_j f(x_j) = \sum_{j=1}^N \delta_{ij} w_j = w_i > 0.$$

Since both $\omega(x)$ and $f(x)$ are non-negative functions, it follows that $w_i > 0$.

Computation of Gaussian quadrature rules

There are many algorithms for computing the nodes x_i and weights w_i of Gaussian quadrature rules. The most popular are the Golub-Welsch algorithm requiring $O(n^2)$ operations, Newton's method for solving $p_n(x) = 0$ using the three-term recurrence for evaluation requiring $O(n^2)$ operations, and asymptotic formulas for large n requiring $O(n)$ operations.

Recurrence relation

Orthogonal polynomials p_r with $(p_r, p_s) = 0$ for $r \neq s$ for a scalar product $(,)$, degree $(p_r) = r$ and leading coefficient one (i.e. monic orthogonal polynomials) satisfy the recurrence relation

$$p_{r+1}(x) = (x - a_{r,r})p_r(x) - a_{r,r-1}p_{r-1}(x) \cdots - a_{r,0}p_0(x)$$

and scalar product defined

$$(f(x), g(x)) = \int_a^b \omega(x) f(x) g(x) dx$$

for $r = 0, 1, \dots, n - 1$ where n is the maximal degree which can be taken to be infinity, and where $a_{r,s} = \frac{(xp_r, p_s)}{(p_s, p_s)}$. First of all, the polynomials defined by the recurrence relation starting with $p_0(x) = 1$ have leading coefficient one and correct degree. Given the starting point by p_0 , the orthogonality of p_r can be shown by induction. For $r = s = 0$ one has

$$(p_1, p_0) = (x - a_{0,0})(p_0, p_0) = (xp_0, p_0) - a_{0,0}(p_0, p_0) = (xp_0, p_0) - (xp_0, p_0) = 0.$$

Now if $p_0, p_1, \dots, p_r$ are orthogonal, then also p_{r+1} , because in

$$(p_{r+1}, p_s) = (xp_r, p_s) - a_{r,r}(p_r, p_s) - a_{r,r-1}(p_{r-1}, p_s) \cdots - a_{r,0}(p_0, p_s)$$

all scalar products vanish except for the first one and the one where p_s meets the same orthogonal polynomial. Therefore,

$$(p_{r+1}, p_s) = (xp_r, p_s) - a_{r,s}(p_s, p_s) = (xp_r, p_s) - (xp_r, p_s) = 0.$$

However, if the scalar product satisfies $(xf, g) = (f, xg)$ (which is the case for Gaussian quadrature), the recurrence relation reduces to a three-term recurrence relation: For $s < r - 1$, xp_s is a polynomial of degree less than or equal to $r - 1$. On the other hand, p_r is orthogonal to every polynomial of degree less than or equal to $r - 1$. Therefore, one has $(xp_r, p_s) = (p_r, xp_s) = 0$ and $a_{r,s} = 0$ for $s < r - 1$. The recurrence relation then simplifies to

$$p_{r+1}(x) = (x - a_{r,r})p_r(x) - a_{r,r-1}p_{r-1}(x)$$

or

$$p_{r+1}(x) = (x - a_r)p_r(x) - b_r p_{r-1}(x)$$

(with the convention $p_{-1}(x) \equiv 0$) where

$$a_r := \frac{(xp_r, p_r)}{(p_r, p_r)}, \quad b_r := \frac{(xp_r, p_{r-1})}{(p_{r-1}, p_{r-1})} = \frac{(p_r, p_r)}{(p_{r-1}, p_{r-1})}$$

(the last because of $(xp_r, p_{r-1}) = (p_r, xp_{r-1}) = (p_r, p_r)$, since xp_{r-1} differs from p_r by a degree less than r).

The Golub-Welsch algorithm

The three-term recurrence relation can be written in matrix form $J\tilde{P} = x\tilde{P} - p_n(x) \times \mathbf{e}_n$ where $\tilde{P} = [p_0(x) \ p_1(x) \ \dots \ p_{n-1}(x)]^T$, $\mathbf{e}_n$ is the n th standard basis vector, i.e., $\mathbf{e}_n = [0 \ \dots \ 0 \ 1]^T$, and J is the so-called Jacobi matrix:

$$J = \begin{pmatrix} a_0 & 1 & 0 & \dots & \dots & \dots \\ b_1 & a_1 & 1 & 0 & \dots & \dots \\ 0 & b_2 & a_2 & 1 & 0 & \dots \\ 0 & \dots & \dots & \dots & \dots & 0 \\ \dots & \dots & 0 & b_{n-2} & a_{n-2} & 1 \\ \dots & \dots & \dots & 0 & b_{n-1} & a_{n-1} \end{pmatrix}$$

The zeros x_j of the polynomials up to degree n , which are used as nodes for the Gaussian quadrature can be found by computing the eigenvalues of this tridiagonal matrix. This procedure is known as *Golub–Welsch algorithm*.

For computing the weights and nodes, it is preferable to consider the symmetric tridiagonal matrix $\mathcal{J}$ with elements

$$\begin{aligned} \mathcal{J}_{i,i} &= J_{i,i} = a_{i-1} & i &= 1, \dots, n \\ \mathcal{J}_{i-1,i} &= \mathcal{J}_{i,i-1} = \sqrt{J_{i,i-1}J_{i-1,i}} = \sqrt{b_{i-1}} & i &= 2, \dots, n. \end{aligned}$$

J and $\mathcal{J}$ are similar matrices and therefore have the same eigenvalues (the nodes). The weights can be computed from the corresponding eigenvectors: If $\phi^{(j)}$ is a normalized eigenvector (i.e., an eigenvector with euclidean norm equal to one) associated to the eigenvalue x_j , the corresponding weight can be computed from the first component of this eigenvector, namely:

$$w_j = \mu_0 \left(\phi_1^{(j)} \right)^2$$

where μ_0 is the integral of the weight function

$$\mu_0 = \int_a^b \omega(x) dx.$$

See, for instance, (Gil, Segura & Temme 2007) for further details.

Error estimates

The error of a Gaussian quadrature rule can be stated as follows (Stoer & Bulirsch 2002, Thm 3.6.24). For an integrand which has $2n$ continuous derivatives,

$$\int_a^b \omega(x) f(x) dx - \sum_{i=1}^n w_i f(x_i) = \frac{f^{(2n)}(\xi)}{(2n)!} (p_n, p_n)$$

for some ξ in (a, b) , where p_n is the monic (i.e. the leading coefficient is 1) orthogonal polynomial of degree n and where

$$(f, g) = \int_a^b \omega(x) f(x) g(x) dx.$$

In the important special case of $\omega(x) = 1$, we have the error estimate (Kahaner, Moler & Nash 1989, §5.2)

$$\frac{(b-a)^{2n+1} (n!)^4}{(2n+1)[(2n)!]^3} f^{(2n)}(\xi), \quad a < \xi < b.$$

Stoer and Bulirsch remark that this error estimate is inconvenient in practice, since it may be difficult to estimate the order $2n$ derivative, and furthermore the actual error may be much less than a bound established by the derivative. Another approach is to use two Gaussian quadrature rules of different orders, and to estimate the error as the difference between the two results. For this purpose, Gauss–Kronrod quadrature rules can be useful.

Gauss–Kronrod rules

If the interval $[a, b]$ is subdivided, the Gauss evaluation points of the new subintervals never coincide with the previous evaluation points (except at zero for odd numbers), and thus the integrand must be evaluated at every point. *Gauss–Kronrod rules* are extensions of Gauss quadrature rules generated by adding $n + 1$ points to an n -point rule in such a way that the resulting rule is of order $2n + 1$. This allows for computing higher-order estimates while re-using the function values of a lower-order estimate. The difference between a Gauss quadrature rule and its Kronrod extension is often used as an estimate of the approximation error.

Gauss–Lobatto rules

Also known as **Lobatto quadrature** (Abramowitz & Stegun 1972, p. 888), named after Dutch mathematician Rehuel Lobatto. It is similar to Gaussian quadrature with the following differences:

1. The integration points include the end points of the integration interval.
2. It is accurate for polynomials up to degree $2n - 3$, where n is the number of integration points (Quarteroni, Sacco & Saleri 2000).

Lobatto quadrature of function $f(x)$ on interval $[-1, 1]$:

$$\int_{-1}^1 f(x) dx = \frac{2}{n(n-1)} [f(1) + f(-1)] + \sum_{i=2}^{n-1} w_i f(x_i) + R_n.$$

Abscissas: x_i is the $(i - 1)$ st zero of $P'_{n-1}(x)$.

Weights:

$$w_i = \frac{2}{n(n-1)[P'_{n-1}(x_i)]^2}, \quad x_i \neq \pm 1.$$

Remainder:

$$R_n = \frac{-n(n-1)^3 2^{2n-1} [(n-2)!]^4}{(2n-1)[(2n-2)!]^3} f^{(2n-2)}(\xi), \quad -1 < \xi < 1.$$

Some of the weights are:

Number of points, n	Points, x_i	Weights, w_i
3	0	$\frac{4}{3}$
	± 1	$\frac{1}{3}$
4	$\pm\sqrt{\frac{1}{5}}$	$\frac{5}{6}$
	± 1	$\frac{1}{6}$
5	0	$\frac{32}{45}$
	$\pm\sqrt{\frac{3}{7}}$	$\frac{49}{90}$
	± 1	$\frac{1}{10}$
6	$\pm\sqrt{\frac{1}{3} - \frac{2\sqrt{7}}{21}}$	$\frac{14 + \sqrt{7}}{30}$
	$\pm\sqrt{\frac{1}{3} + \frac{2\sqrt{7}}{21}}$	$\frac{14 - \sqrt{7}}{30}$
	± 1	$\frac{1}{15}$
7	0	$\frac{256}{525}$
	$\pm\sqrt{\frac{5}{11} - \frac{2}{11}\sqrt{\frac{5}{3}}}$	$\frac{124 + 7\sqrt{15}}{350}$
	$\pm\sqrt{\frac{5}{11} + \frac{2}{11}\sqrt{\frac{5}{3}}}$	$\frac{124 - 7\sqrt{15}}{350}$
	± 1	$\frac{1}{21}$

An adaptive variant of this algorithm with 2 interior nodes^[3] is found in GNU Octave and MATLAB as `quad1` and `integrate`.^{[4][5]}

References

- Implementing an Accurate Generalized Gaussian Quadrature Solution to Find the Elastic Field in a Homogeneous Anisotropic Media (http://www.sscm.kg.ac.rs/jsscm/downloads/Vol11No1/Vol11No1_02.pdf)
- Abramowitz, Milton; Stegun, Irene Ann, eds. (1983) [June 1964]. "Chapter 25.4, Integration". *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Applied Mathematics Series. **55** (Ninth reprint with additional corrections of tenth original printing with corrections (December 1972); first ed.). Washington D.C.; New York: United States Department of Commerce, National Bureau of Standards; Dover Publications. ISBN 978-0-486-61272-0. LCCN 64-60036 (<http://lccn.loc.gov/64-60036>). MR 0167642 (<https://www.ams.org/mathscinet-getitem?mr=0167642>). LCCN 65-12253 (<https://lccn.loc.gov/65-12253>).
- Anderson, Donald G. (1965). "Gaussian quadrature formulae for $\int_0^1 -\ln(x)f(x)dx$ " (<https://doi.org/10.1090/s0025-5718-1965-0178569-1>). *Math. Comp.* **19** (91): 477–481. doi:10.1090/s0025-5718-1965-0178569-1 (<https://doi.org/10.1090%2Fs0025-5718-1965-0178569-1>).
- Golub, Gene H.; Welsch, John H. (1969), "Calculation of Gauss Quadrature Rules", *Mathematics of Computation*, **23** (106): 221–230, doi:10.1090/S0025-5718-69-99647-1 (<https://doi.org/10.1090%2FS0025-5718-69-99647-1>), JSTOR 2004418 (<https://www.jstor.org/stable/2004418>)
- Gautschi, Walter (1968). "Construction of Gauss–Christoffel Quadrature Formulas". *Math. Comp.* **22** (102). pp. 251–270. doi:10.1090/S0025-5718-1968-0228171-0 (<https://doi.org/10.1090%2FS0025-5718-1968-0228171-0>). MR 0228171 (<https://www.ams.org/mathscinet-getitem?mr=0228171>).
- Gautschi, Walter (1970). "On the construction of Gaussian quadrature rules from modified moments". *Math. Comp.* **24**. pp. 245–260. doi:10.1090/S0025-5718-1970-0285117-6 (<https://doi.org/10.1090%2FS0025-5718-1970-0285117-6>). MR 0285177 (<https://www.ams.org/mathscinet-getitem?mr=0285177>).
- Piessens, R. (1971). "Gaussian quadrature formulas for the numerical integration of Bromwich's integral and the inversion of the laplace transform". *J. Eng. Math.* **5** (1). pp. 1–9. Bibcode:1971JEnMa...5....1P (<https://ui.adsabs.harvard.edu/abs/1971JEnMa...5....1P>). doi:10.1007/BF01535429 (<https://doi.org/10.1007%2FBF01535429>).

- Danloy, Bernard (1973). "Numerical construction of Gaussian quadrature formulas for $\int_0^1 (-\log x)x^\alpha f(x)dx$ and $\int_0^\infty E_m(x)f(x)dx$ ". *Math. Comp.* **27** (124). pp. 861–869. doi:10.1090/S0025-5718-1973-0331730-X (https://doi.org/10.1090/S0025-5718-1973-0331730-X). MR 0331730 (https://www.ams.org/mathscinet-getitem?mr=0331730).
- Kahaner, David; Moler, Cleve; Nash, Stephen (1989), *Numerical Methods and Software* (https://archive.org/details/numericalmethods0000kaha), Prentice-Hall, ISBN 978-0-13-627258-8
- Sagar, Robin P. (1991). "A Gaussian quadrature for the calculation of generalized Fermi-Dirac integrals". *Comput. Phys. Commun.* **66** (2–3): 271–275. Bibcode:1991CoPhC..66..271S (https://ui.adsabs.harvard.edu/abs/1991CoPhC..66..271S). doi:10.1016/0010-4655(91)90076-W (https://doi.org/10.1016/0010-4655(91)90076-W).
- Yakimiw, E. (1996). "Accurate computation of weights in classical Gauss-Christoffel quadrature rules". *J. Comput. Phys.* **129** (2): 406–430. Bibcode:1996JCoPh.129..406Y (https://ui.adsabs.harvard.edu/abs/1996JCoPh.129..406Y). doi:10.1006/jcph.1996.0258 (https://doi.org/10.1006/jcph.1996.0258).
- Laurie, Dirk P. (1999). "Accurate recovery of recursion coefficients from Gaussian quadrature formulas", *J. Comput. Appl. Math.*, **112** (1–2): 165–180, doi:10.1016/S0377-0427(99)00228-9 (https://doi.org/10.1016/S0377-0427(99)00228-9)
- Laurie, Dirk P. (2001). "Computation of Gauss-type quadrature formulas" (https://doi.org/10.1016/S0377-0427(00)00506-9). *J. Comput. Appl. Math.* **127** (1–2): 201–217. Bibcode:2001JCoAM.127..201L (https://ui.adsabs.harvard.edu/abs/2001JCoAM.127..201L). doi:10.1016/S0377-0427(00)00506-9 (https://doi.org/10.1016/S0377-0427(00)00506-9).
- Riener, Cordian; Schweighofer, Markus (2018). "Optimization approaches to quadrature: New characterizations of Gaussian quadrature on the line and quadrature with few nodes on plane algebraic curves, on the plane and in higher dimensions". *Journal of Complexity*. **45**: 22–54. arXiv:1607.08404 (https://arxiv.org/abs/1607.08404). doi:10.1016/j.jco.2017.10.002 (https://doi.org/10.1016/j.jco.2017.10.002).
- Stoer, Josef; Bulirsch, Roland (2002), *Introduction to Numerical Analysis* (3rd ed.), Springer, ISBN 978-0-387-95452-3.
- Temme, Nico M. (2010), "§3.5(v): Gauss Quadrature" (http://dlmf.nist.gov/3.5.v), in Olver, Frank W. J.; Lozier, Daniel M.; Boisvert, Ronald F.; Clark, Charles W. (eds.), *NIST Handbook of Mathematical Functions*, Cambridge University Press, ISBN 978-0-521-19225-5, MR 2723248 (https://www.ams.org/mathscinet-getitem?mr=2723248)
- Press, WH; Teukolsky, SA; Vetterling, WT; Flannery, BP (2007), "Section 4.6. Gaussian Quadratures and Orthogonal Polynomials" (http://apps.nrbook.com/empanel/index.html?pg=179), *Numerical Recipes: The Art of Scientific Computing* (3rd ed.), New York: Cambridge University Press, ISBN 978-0-521-88068-8
- Gil, Amparo; Segura, Javier; Temme, Nico M. (2007), "§5.3: Gauss quadrature", *Numerical Methods for Special Functions*, SIAM, ISBN 978-0-89871-634-4
- Quarteroni, Alfio; Sacco, Riccardo; Saleri, Fausto (2000). *Numerical Mathematics*. New York: Springer-Verlag. pp. 422 (https://archive.org/details/numericalmathema00quar_579/page/n439), 425. ISBN 0-387-98959-5.

Specific

1. *Methodus nova integralium valores per approximationem inveniendi*. In: *Comm. Soc. Sci. Göttingen Math.* Band 3, 1815, S. 29–76, Gallica (http://gallica.bnf.fr/ark:/12148/bpt6k2412190.r=Gauss.langEN), datiert 1814, auch in Werke, Band 3, 1876, S. 163–196.
2. C. G. J. Jacobi: *Ueber Gauß' neue Methode, die Werthe der Integrale näherungsweise zu finden*. In: *Journal für Reine und Angewandte Mathematik*. Band 1, 1826, S. 301–308, (online) (http://gdz.sub.uni-goettingen.de/dms/load/img/?PPN=PPN243919689_0001&DMDID=DMDLOG_0035), und Werke, Band 6.
3. Gander, Walter; Gautschi, Walter (2000). "Adaptive Quadrature - Revisited" (https://www.inf.ethz.ch/personal/gander/). *BIT Numerical Mathematics*. **40** (1): 84–101. doi:10.1023/A:1022318402393 (https://doi.org/10.1023/A:1022318402393).
4. "Numerical integration - MATLAB integral" (https://www.mathworks.com/help/matlab/ref/integral.html).
5. "Functions of One Variable (GNU Octave)" (https://octave.org/doc/v4.2.2/Functions-of-One-Variable.html#XREFquadl). Retrieved 28 September 2018.

External links

- "Gauss quadrature formula" (https://www.encyclopediaofmath.org/index.php?title=Gauss_quadrature_formula), *Encyclopedia of Mathematics*, EMS Press, 2001 [1994]
- ALGLIB (http://www.alglib.net/integral/gq/) contains a collection of algorithms for numerical integration (in C# / C++ / Delphi / Visual Basic / etc.)
- GNU Scientific Library (https://www.gnu.org/software/gsl/) — includes C version of QUADPACK algorithms (see also GNU Scientific Library)
- From Lobatto Quadrature to the Euler constant e (http://ans.hsh.no/home/skk/Publications/Lobatto/PRIMUS_KHATTRI.pdf)
- Gaussian Quadrature Rule of Integration – Notes, PPT, Matlab, Mathematica, Maple, Mathcad (http://numericalmethods.eng.usf.edu/topics/gauss_quadrature.html) at *Holistic Numerical Methods Institute*
- Weisstein, Eric W. "Legendre-Gauss Quadrature" (https://mathworld.wolfram.com/Legendre-GaussQuadrature.html). *MathWorld*.
- Gaussian Quadrature (http://demonstrations.wolfram.com/GaussianQuadrature/) by Chris Maes and Anton Antonov, *Wolfram Demonstrations Project*.
- Tabulated weights and abscissae with Mathematica source code (https://pomax.github.io/bezierinfo/legendre-gauss.html), high precision (16 and 256 decimal places) Legendre-Gaussian quadrature weights and abscissas, for $n=2$ through $n=64$, with Mathematica source code.
- Mathematica source code distributed under the GNU LGPL (https://web.archive.org/web/20121122202131/http://people.sc.fsu.edu/~jburkardt/math_src/arbitrary_weight_rule/arbitrary_weight_rule.html) for abscissas and weights generation for arbitrary weighting functions $W(x)$, integration domains and precisions.

- [Gaussian Quadrature in Boost.Math, for arbitrary precision and approximation order \(http://www.boost.org/doc/libs/release/libs/math/doc/html/math_toolkit/gauss.html\)](http://www.boost.org/doc/libs/release/libs/math/doc/html/math_toolkit/gauss.html)
 - [Gauss-Kronrod Quadrature in Boost.Math \(http://www.boost.org/doc/libs/release/libs/math/doc/html/math_toolkit/gauss_kronrod.html\)](http://www.boost.org/doc/libs/release/libs/math/doc/html/math_toolkit/gauss_kronrod.html)
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